

# Track 2 Best Model Collage Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

## Overview

This report compares representative Track 2 TE-curve predictions for the current best reference, Track 1, Wave 1 directional, and Wave 1 global models. Each model is shown as one four-image collage so local oscillation tracking can be inspected directly.

## Scope

- each collage contains four deterministic held-out test curves;
- forward models are shown on forward curves only;
- backward models are shown on backward curves only;
- global Wave 1 models are shown on two forward and two backward curves;
- Measured TE uses the same line width as predictions and a dark-gray color for balanced visual comparison.

## Metrics Summary

### Forward Reference Best Models

| Candidate              | Source        | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|------------------------|---------------|---------|-----------------|------------------|------------|
| paper_original_best_Fw | rcim_original | Fw      | 0.002769        | 0.002951         | 6.250      |
| paper_retuned_best_Fw  | rcim_retuned  | Fw      | 0.001839        | 0.002041         | 4.109      |
| track1_best_Fw         | rcim_track1   | Fw      | 0.003014        | 0.003204         | 6.819      |

### Forward Wave 1 Family Best Models

| Candidate                | Source                               | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|--------------------------|--------------------------------------|---------|-----------------|------------------|------------|
| feedforward_fw           | wave1_current_registry               | Fw      | 0.003404        | 0.003855         | 7.551      |
| harmonic_regression_fw   | wave1_current_registry               | Fw      | 0.003230        | 0.003494         | 7.185      |
| periodic_mlp_fw          | wave1_current_registry               | Fw      | 0.003254        | 0.003553         | 7.232      |
| residual_harmonic_mlp_fw | wave1_current_registry               | Fw      | 0.003273        | 0.003563         | 7.266      |
| tree_fw                  | wave1_current_registry               | Fw      | 0.003053        | 0.003395         | 6.731      |
| periodic_mlp_harmonic_fw | wave1_periodic_mlp_harmonic_campaign | Fw      | 0.003254        | 0.003553         | 7.232      |

### Backward Reference Best Models

| Candidate             | Source       | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|-----------------------|--------------|---------|-----------------|------------------|------------|
| paper_retuned_best_Bw | rcim_retuned | Bw      | 0.003675        | 0.004284         | 7.572      |
| track1_best_Bw        | rcim_track1  | Bw      | 0.005027        | 0.005212         | 11.860     |

### Backward Wave 1 Family Best Models

| Candidate                | Source                               | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|--------------------------|--------------------------------------|---------|-----------------|------------------|------------|
| feedforward_bw           | wave1_current_registry               | Bw      | 0.003586        | 0.004023         | 7.832      |
| harmonic_regression_bw   | wave1_current_registry               | Bw      | 0.003678        | 0.004012         | 8.058      |
| periodic_mlp_bw          | wave1_current_registry               | Bw      | 0.003574        | 0.004006         | 7.807      |
| residual_harmonic_mlp_bw | wave1_current_registry               | Bw      | 0.003536        | 0.003874         | 7.728      |
| tree_bw                  | wave1_current_registry               | Bw      | 0.003258        | 0.003651         | 7.051      |
| periodic_mlp_harmonic_bw | wave1_periodic_mlp_harmonic_campaign | Bw      | 0.003583        | 0.003925         | 7.875      |

### Forward Wave 2 Temporal Family Best Models

| Candidate               | Source                        | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|-------------------------|-------------------------------|---------|-----------------|------------------|------------|
| temporal_convolution_fw | wave2_temporal_entry_registry | Fw      | 0.003603        | 0.004031         | 8.028      |
| gru_sequence_fw         | wave2_temporal_entry_registry | Fw      | 0.003330        | 0.003762         | 7.378      |
| lstm_sequence_fw        | wave2_temporal_entry_registry | Fw      | 0.003366        | 0.003800         | 7.450      |

### Backward Wave 2 Temporal Family Best Models

| Candidate               | Source                        | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|-------------------------|-------------------------------|---------|-----------------|------------------|------------|
| temporal_convolution_bw | wave2_temporal_entry_registry | Bw      | 0.003742        | 0.004166         | 8.184      |
| gru_sequence_bw         | wave2_temporal_entry_registry | Bw      | 0.003626        | 0.004082         | 7.907      |
| lstm_sequence_bw        | wave2_temporal_entry_registry | Bw      | 0.003555        | 0.003985         | 7.767      |

### Global Wave 1 Family Best Models

| Candidate                    | Source                               | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|------------------------------|--------------------------------------|---------|-----------------|------------------|------------|
| feedforward_global           | wave1_current_registry               | global  | 0.003465        | 0.003897         | 7.636      |
| harmonic_regression_global   | wave1_current_registry               | global  | 0.018129        | 0.018330         | 41.458     |
| periodic_mlp_global          | wave1_current_registry               | global  | 0.003447        | 0.003872         | 7.582      |
| residual_harmonic_mlp_global | wave1_current_registry               | global  | 0.003407        | 0.003822         | 7.486      |
| tree_global                  | wave1_current_registry               | global  | 0.003144        | 0.003533         | 6.854      |
| periodic_mlp_harmonic_global | wave1_periodic_mlp_harmonic_campaign | global  | 0.003516        | 0.003810         | 7.779      |

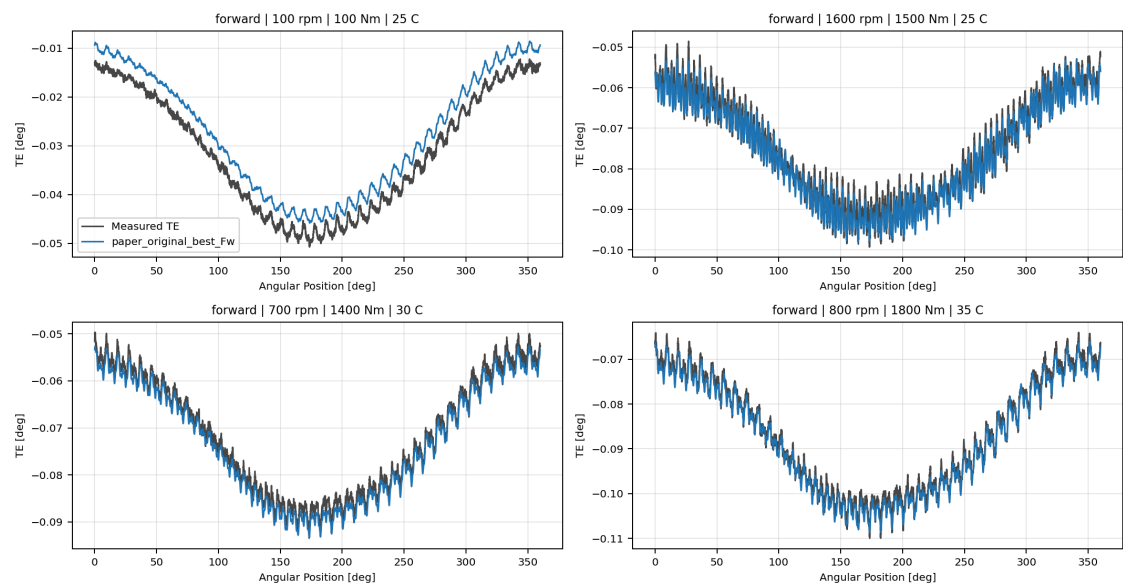
Global Wave 2 Temporal Family Best Models

| Candidate                   | Source                        | Surface | Curve MAE [deg] | Curve RMSE [deg] | Mean Error |
|-----------------------------|-------------------------------|---------|-----------------|------------------|------------|
| temporal_convolution_global | wave2_temporal_entry_registry | global  | 0.003751        | 0.004183         | 8.295      |
| gru_sequence_global         | wave2_temporal_entry_registry | global  | 0.003591        | 0.004028         | 7.907      |
| lstm_sequence_global        | wave2_temporal_entry_registry | global  | 0.003480        | 0.003903         | 7.654      |

# Collage Gallery - Forward Reference Best Models

paper\_original\_best\_Fw

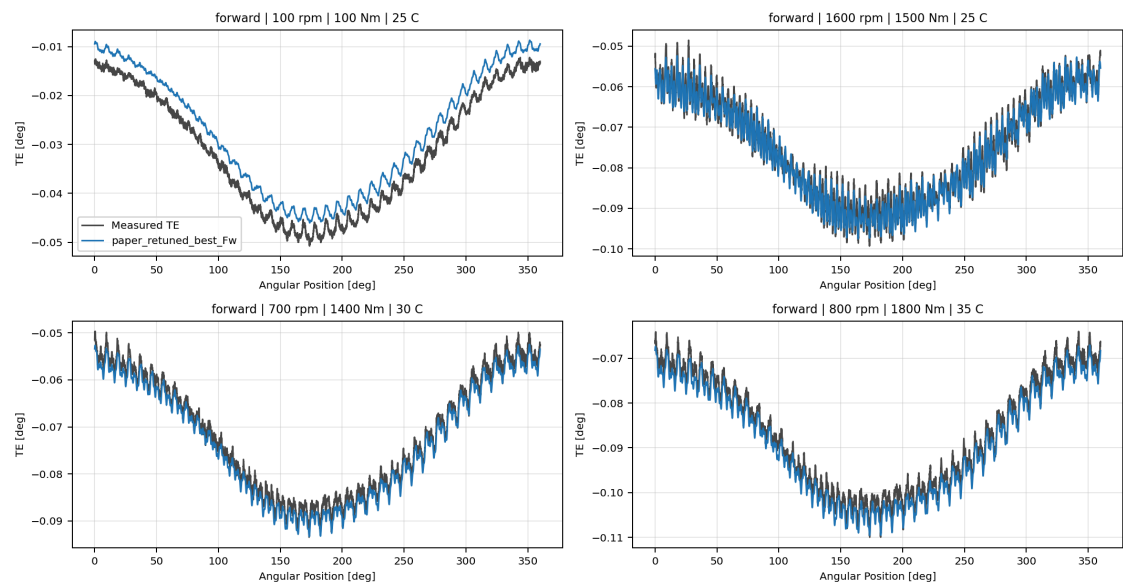
paper\_original\_best\_Fw



paper\_original\_best\_Fw Track 2 collage

paper\_retuned\_best\_Fw

paper\_retuned\_best\_Fw



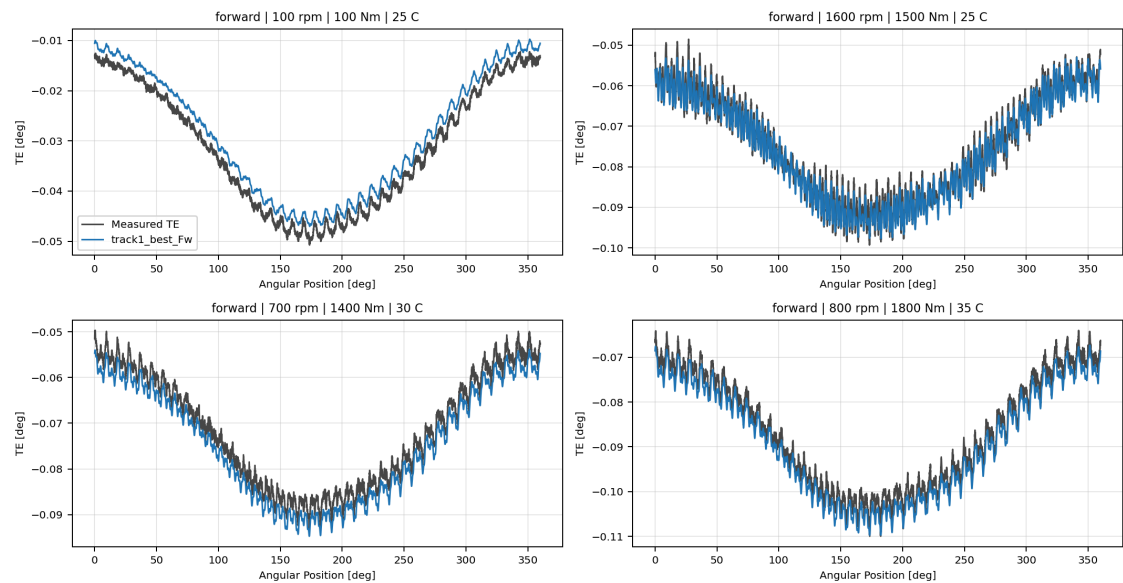
paper\_retuned\_best\_Fw Track 2 collage



# Collage Gallery - Forward Reference Best Models Continued

track1\_best\_Fw

track1\_best\_Fw

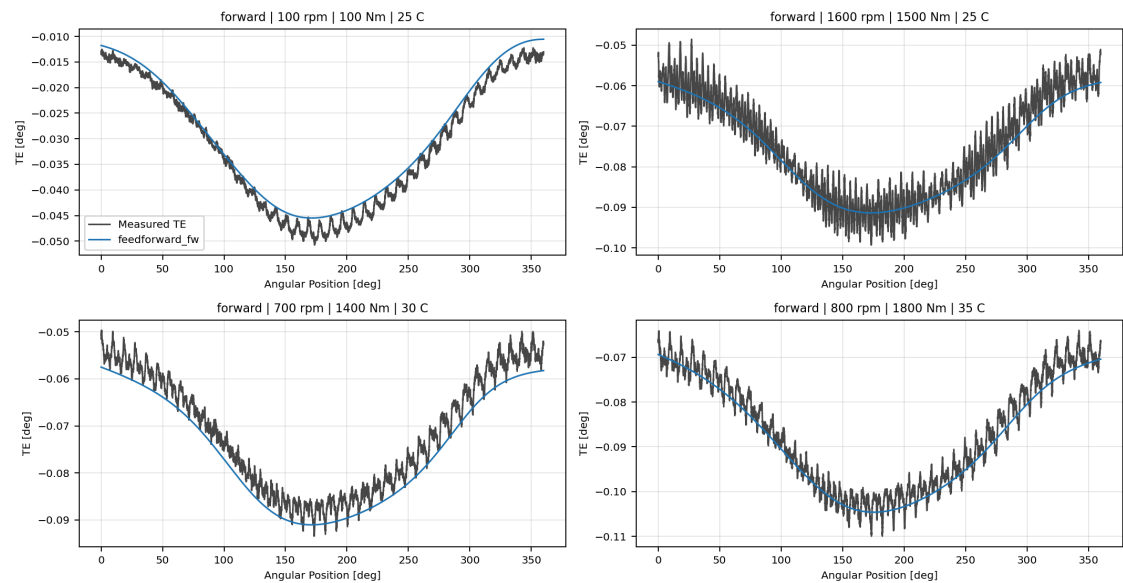


track1\_best\_Fw Track 2 collage

# Collage Gallery - Forward Wave 1 Family Best Models

feedforward\_fw

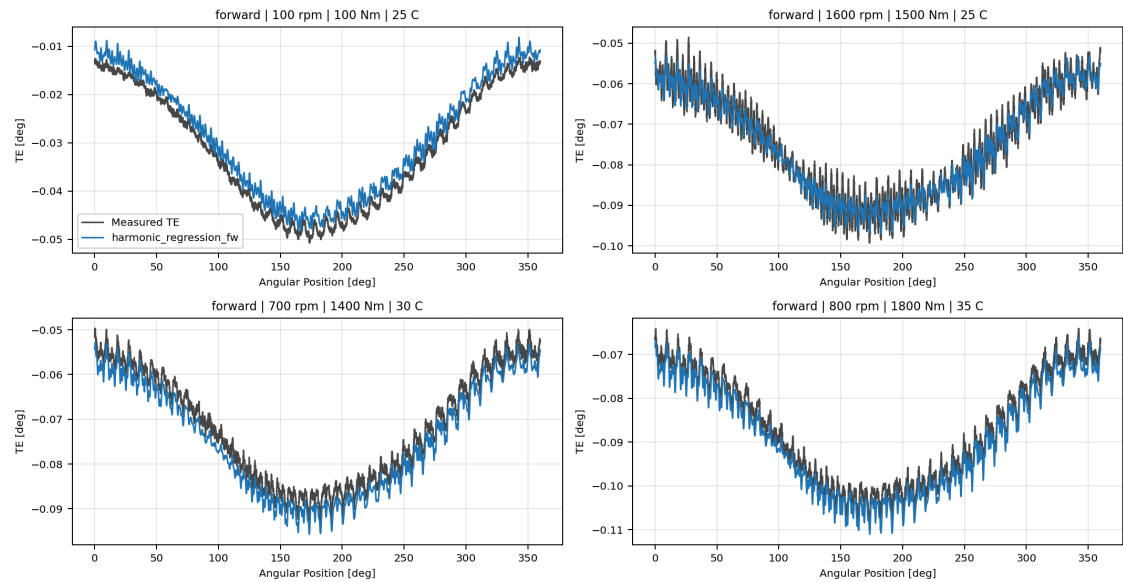
feedforward\_fw



feedforward\_fw Track 2 collage

harmonic\_regression\_fw

harmonic\_regression\_fw

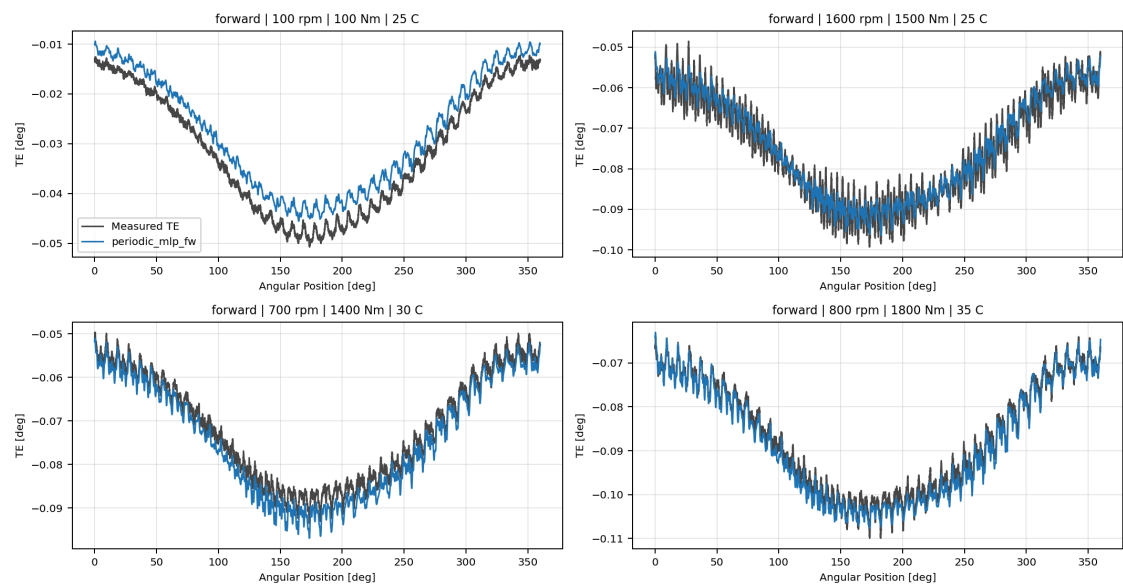


harmonic\_regression\_fw Track 2 collage

# Collage Gallery - Forward Wave 1 Family Best Models Continued

periodic\_mlp\_fw

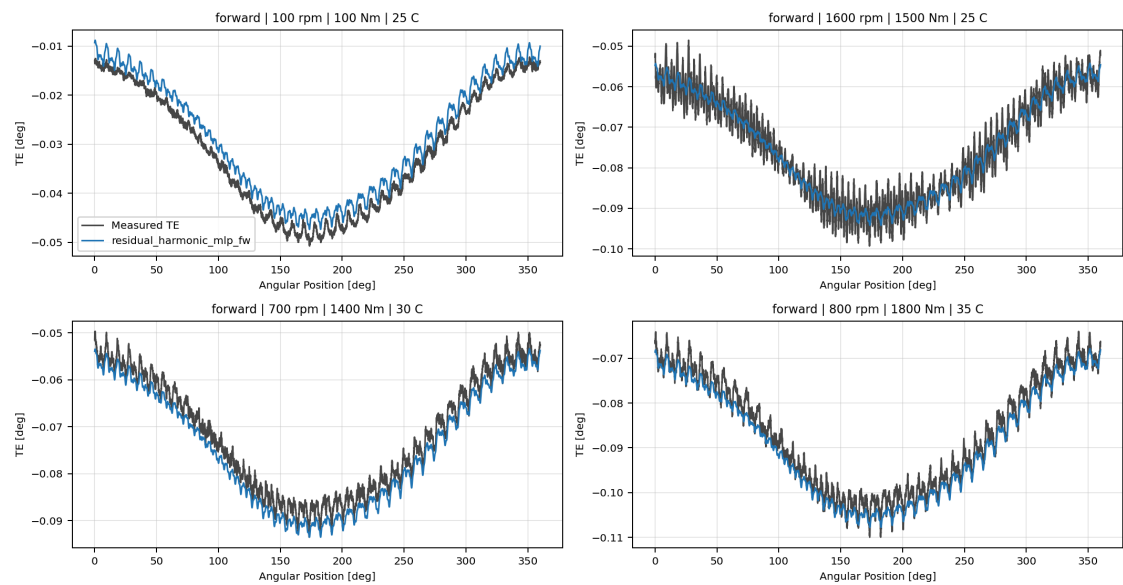
periodic\_mlp\_fw



periodic\_mlp\_fw Track 2 collage

residual\_harmonic\_mlp\_fw

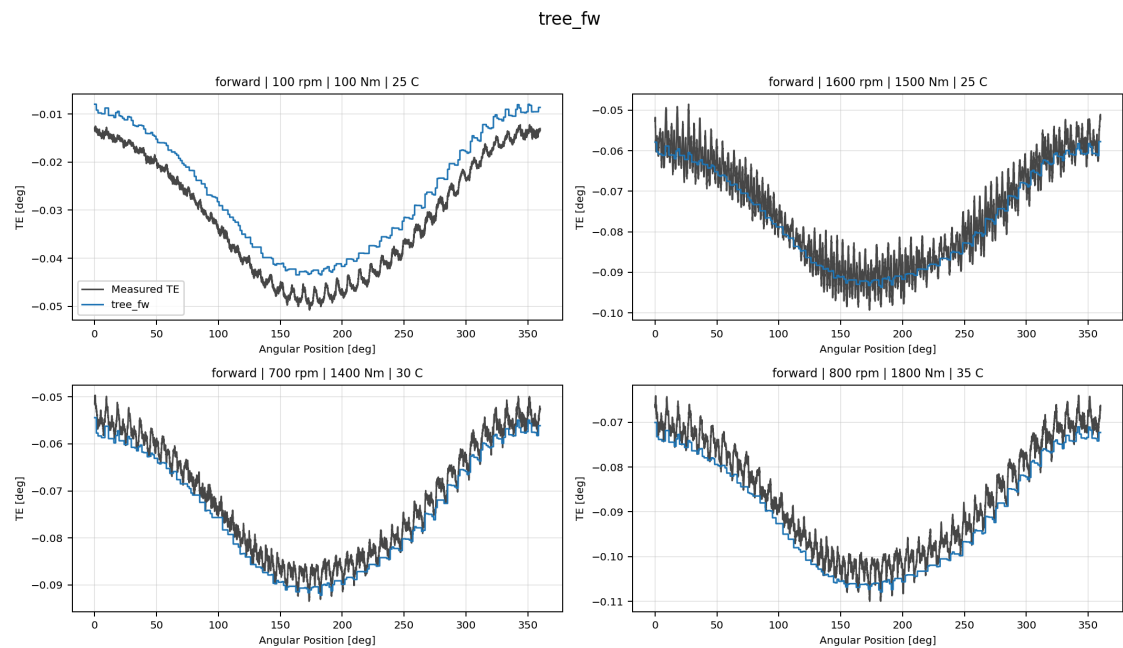
residual\_harmonic\_mlp\_fw



residual\_harmonic\_mlp\_fw Track 2 collage

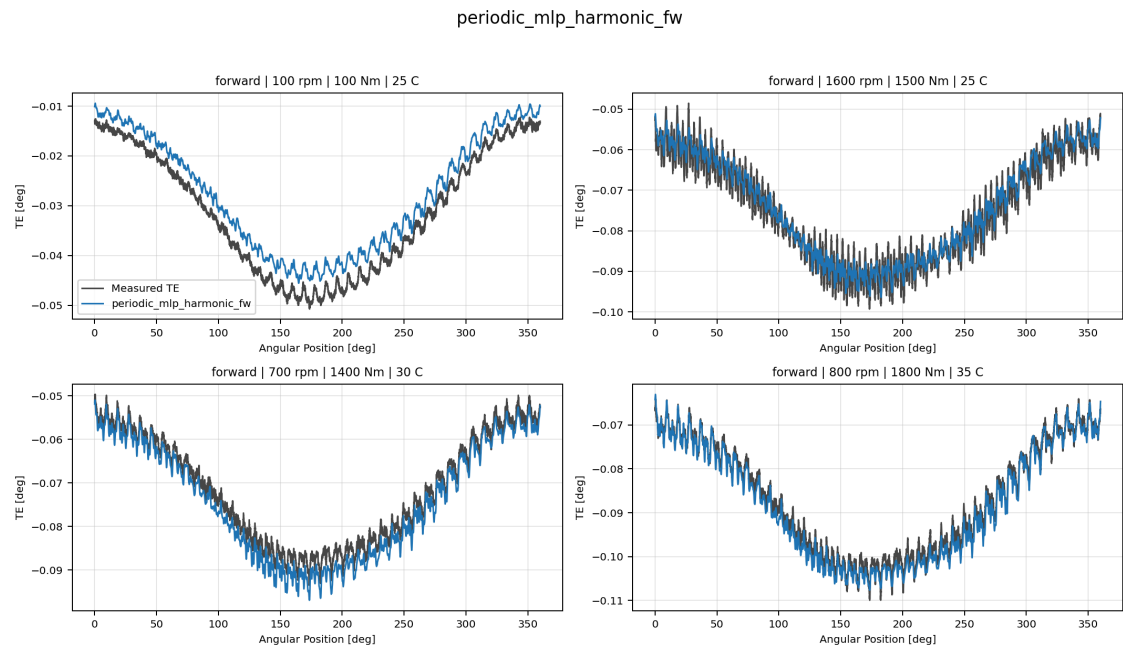
# Collage Gallery - Forward Wave 1 Family Best Models Continued 2

tree\_fw



tree\_fw Track 2 collage

periodic\_mlp\_harmonic\_fw

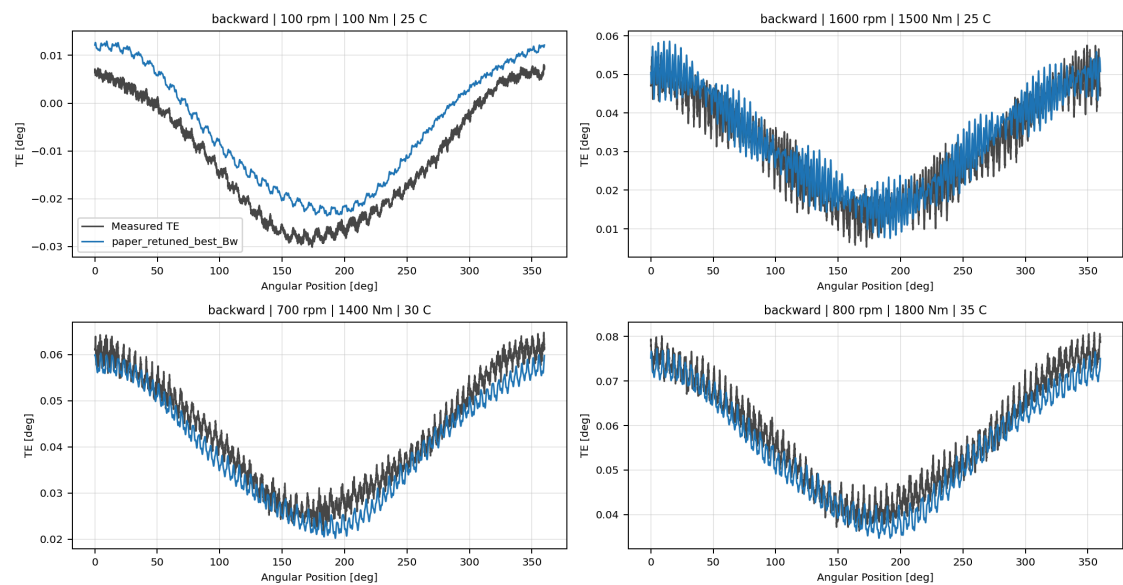


periodic\_mlp\_harmonic\_fw Track 2 collage

# Collage Gallery - Backward Reference Best Models

paper\_retuned\_best\_Bw

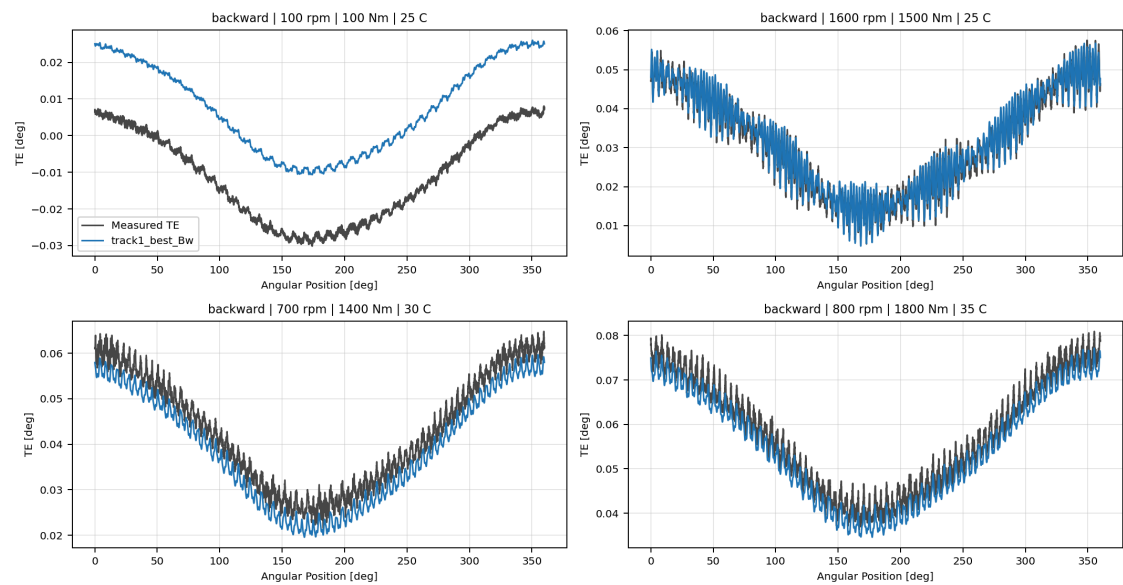
paper\_retuned\_best\_Bw



paper\_retuned\_best\_Bw Track 2 collage

track1\_best\_Bw

track1\_best\_Bw

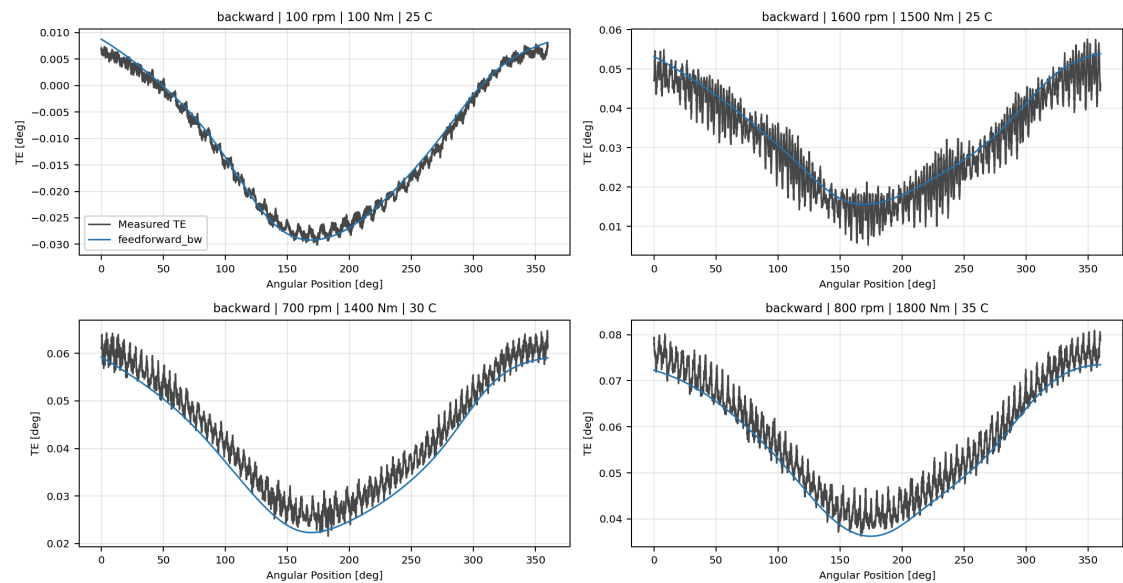


track1\_best\_Bw Track 2 collage

# Collage Gallery - Backward Wave 1 Family Best Models

feedforward\_bw

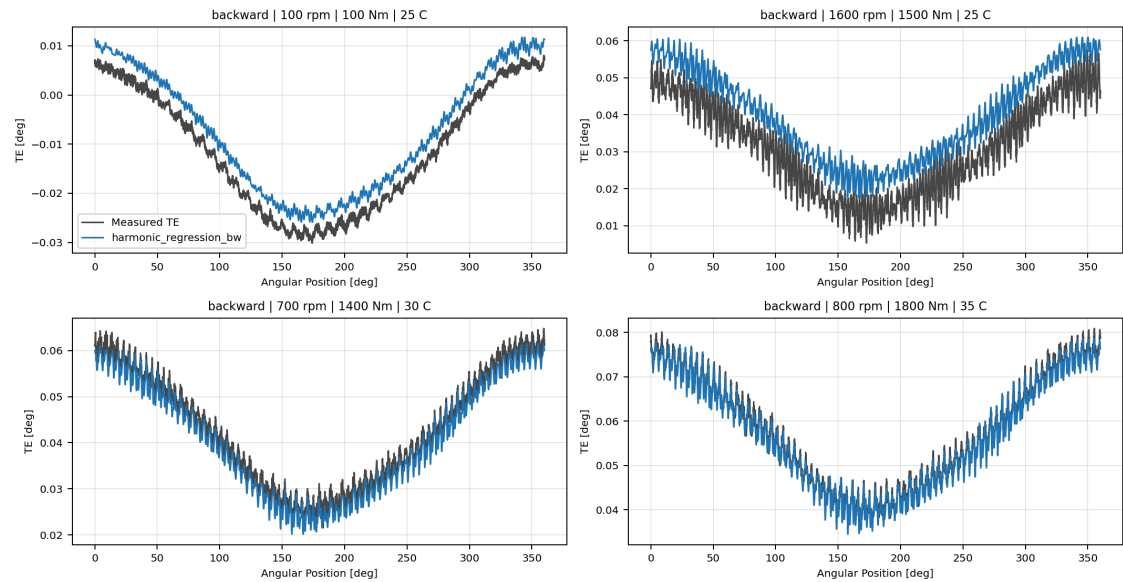
feedforward\_bw



feedforward\_bw Track 2 collage

harmonic\_regression\_bw

harmonic\_regression\_bw

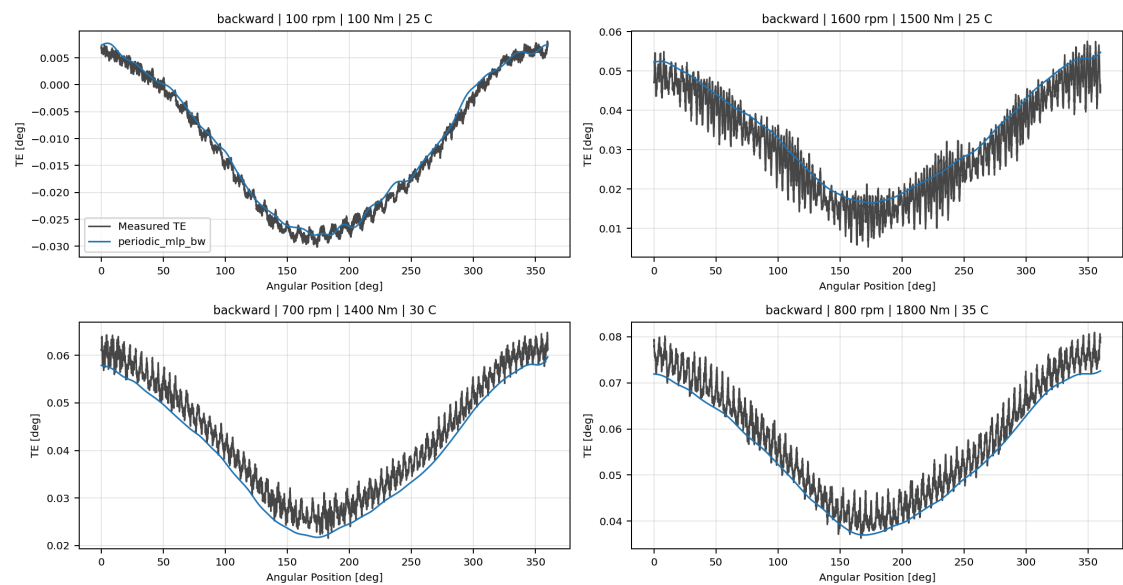


harmonic\_regression\_bw Track 2 collage

# Collage Gallery - Backward Wave 1 Family Best Models Continued

periodic\_mlp\_bw

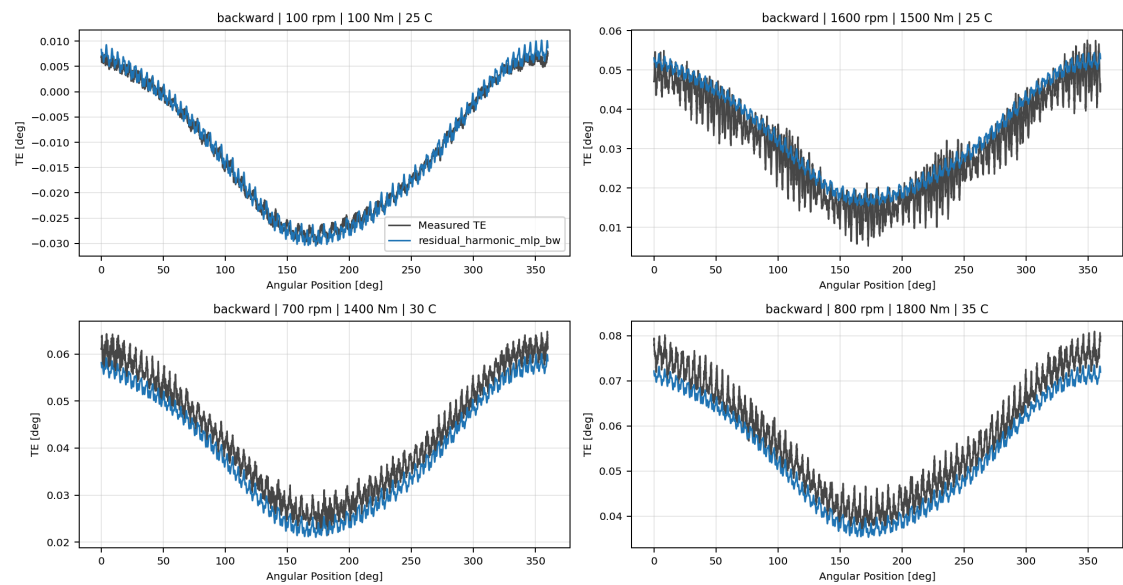
periodic\_mlp\_bw



periodic\_mlp\_bw Track 2 collage

residual\_harmonic\_mlp\_bw

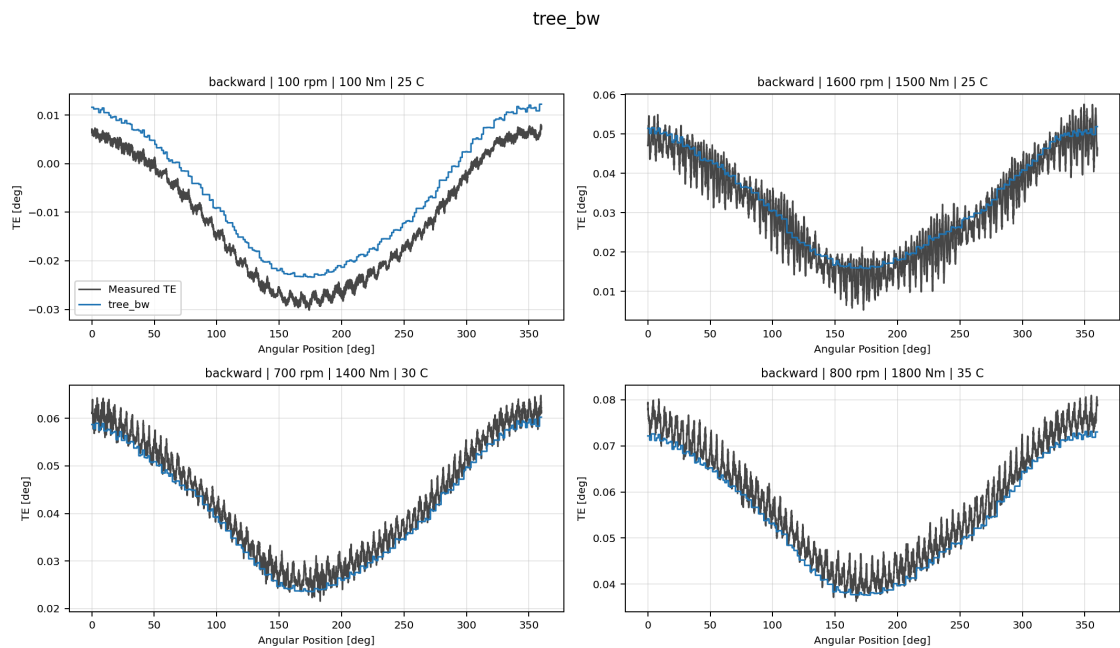
residual\_harmonic\_mlp\_bw



residual\_harmonic\_mlp\_bw Track 2 collage

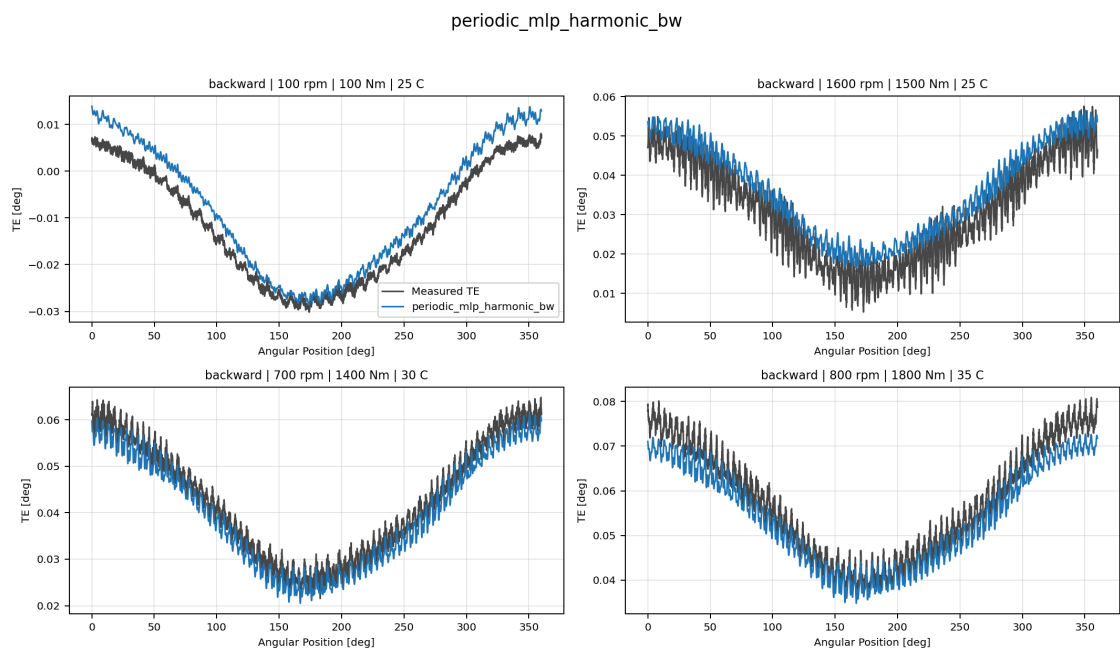
# Collage Gallery - Backward Wave 1 Family Best Models Continued 2

tree\_bw



tree\_bw Track 2 collage

periodic\_mlp\_harmonic\_bw



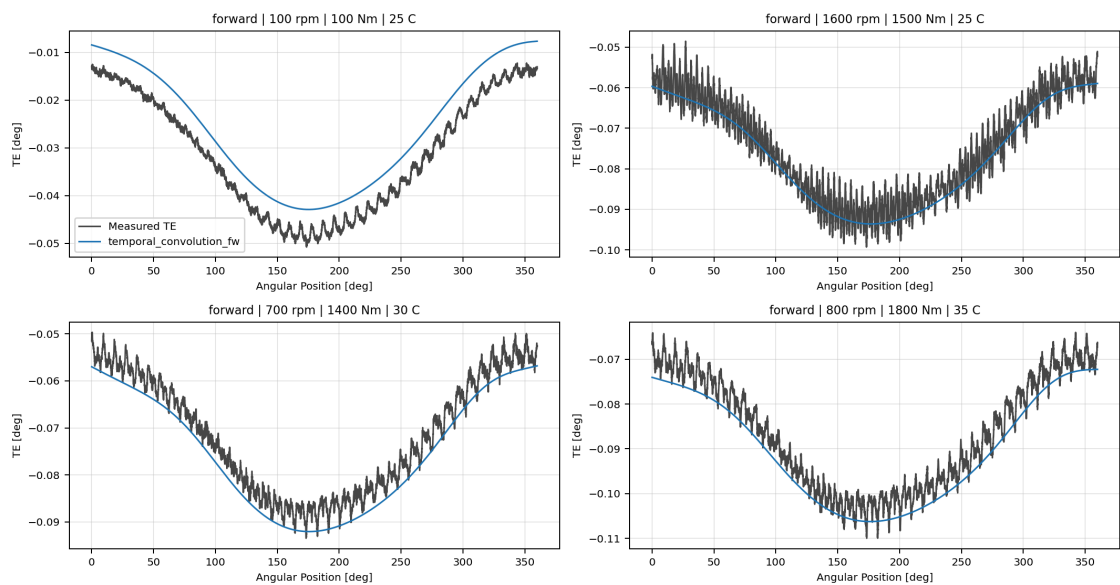
periodic\_mlp\_harmonic\_bw Track 2 collage



# Collage Gallery - Forward Wave 2 Temporal Family Best Models

temporal\_convolution\_fw

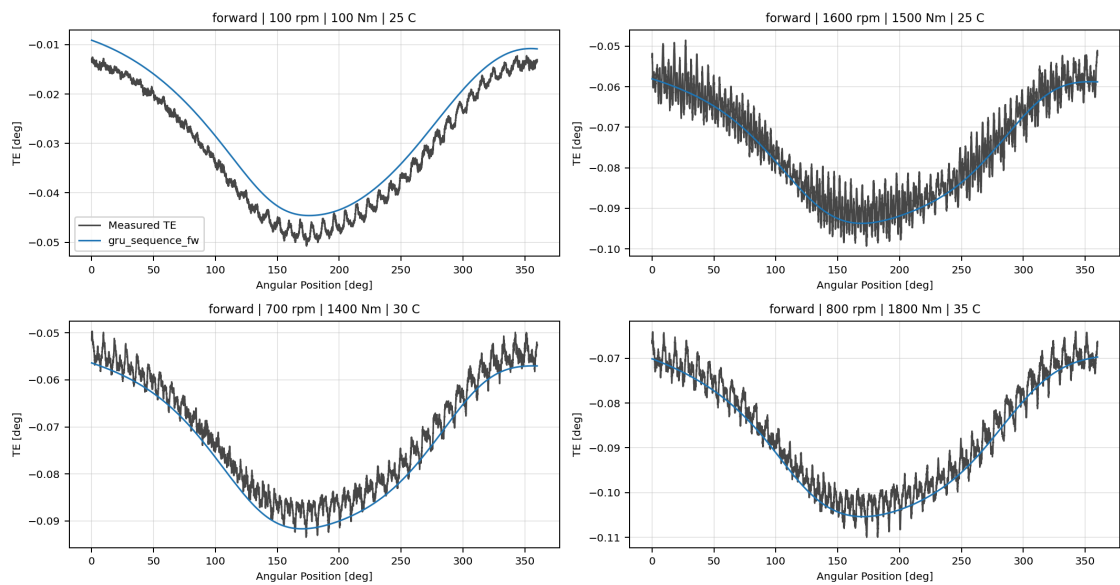
temporal\_convolution\_fw



temporal\_convolution\_fw Track 2 collage

gru\_sequence\_fw

gru\_sequence\_fw

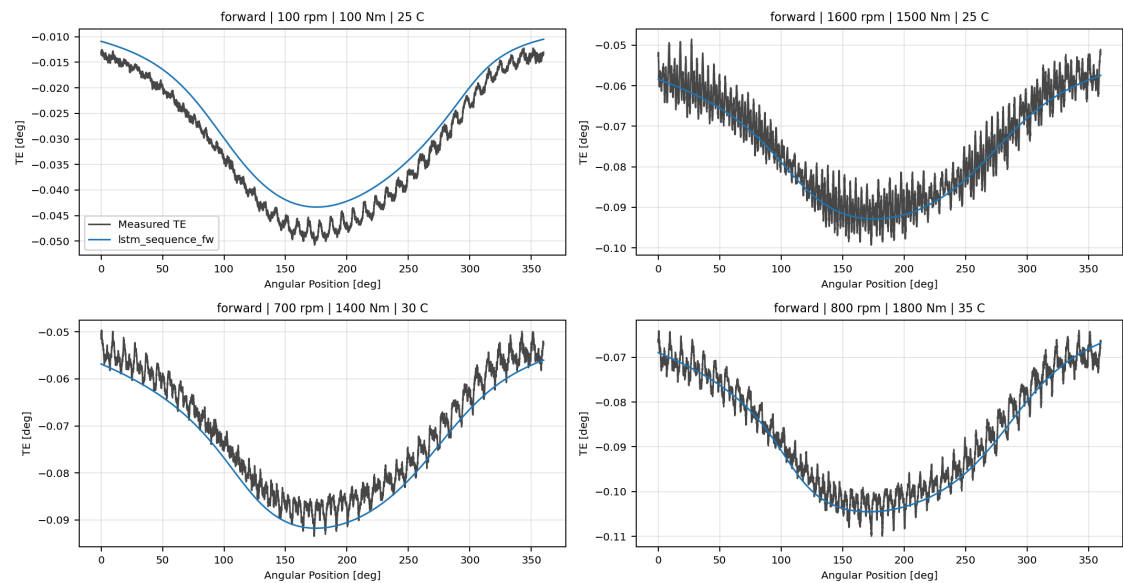


gru\_sequence\_fw Track 2 collage

# Collage Gallery - Forward Wave 2 Temporal Family Best Models Continued

lstm\_sequence\_fw

lstm\_sequence\_fw

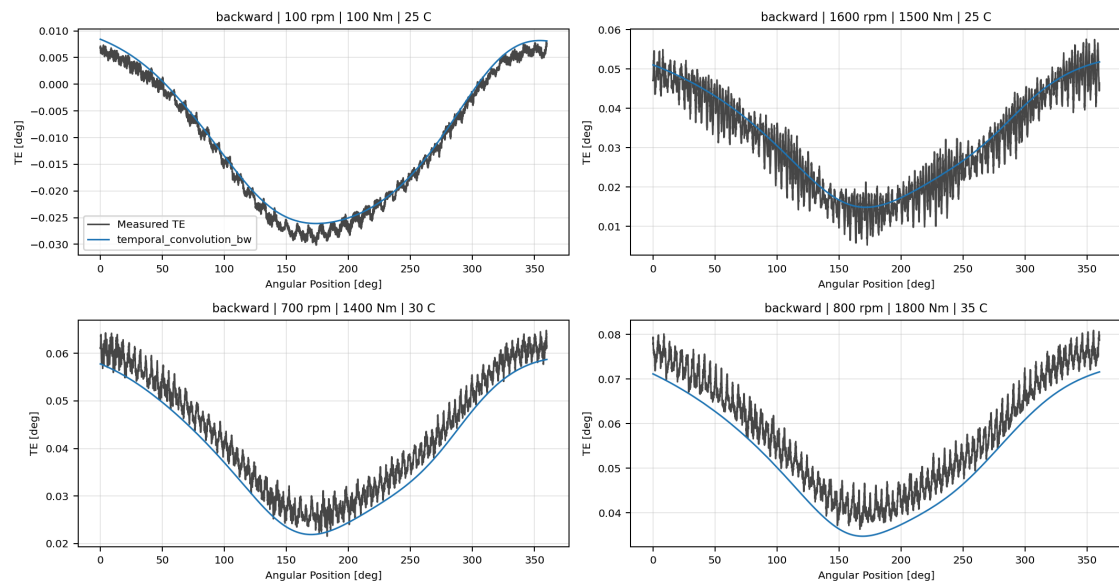


lstm\_sequence\_fw Track 2 collage

## Collage Gallery - Backward Wave 2 Temporal Family Best Models

temporal\_convolution\_bw

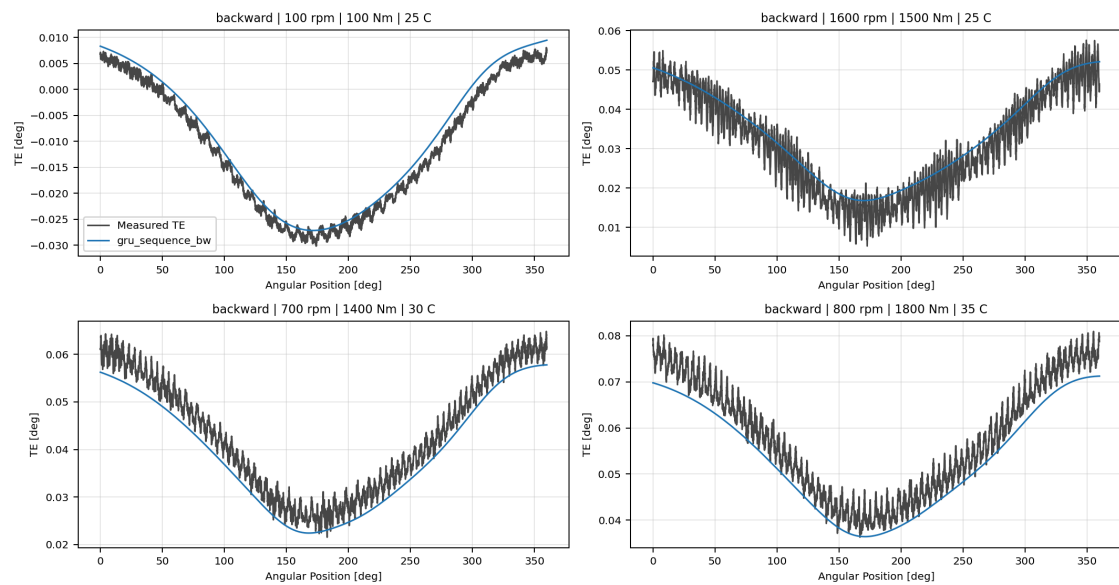
temporal\_convolution\_bw



temporal\_convolution\_bw Track 2 collage

gru\_sequence\_bw

gru\_sequence\_bw

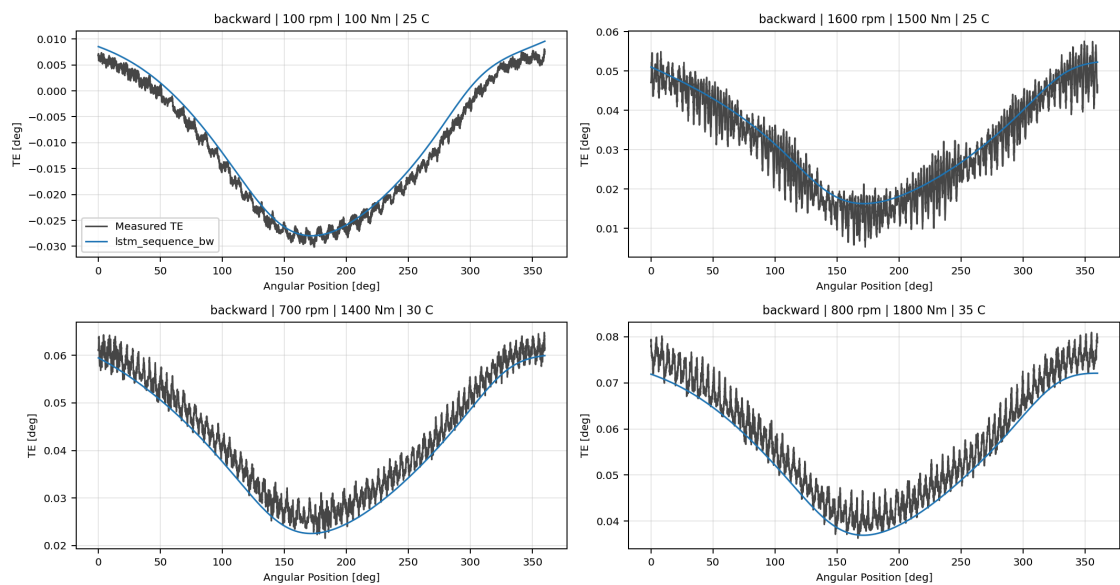


gru\_sequence\_bw Track 2 collage

# Collage Gallery - Backward Wave 2 Temporal Family Best Models Continued

lstm\_sequence\_bw

lstm\_sequence\_bw

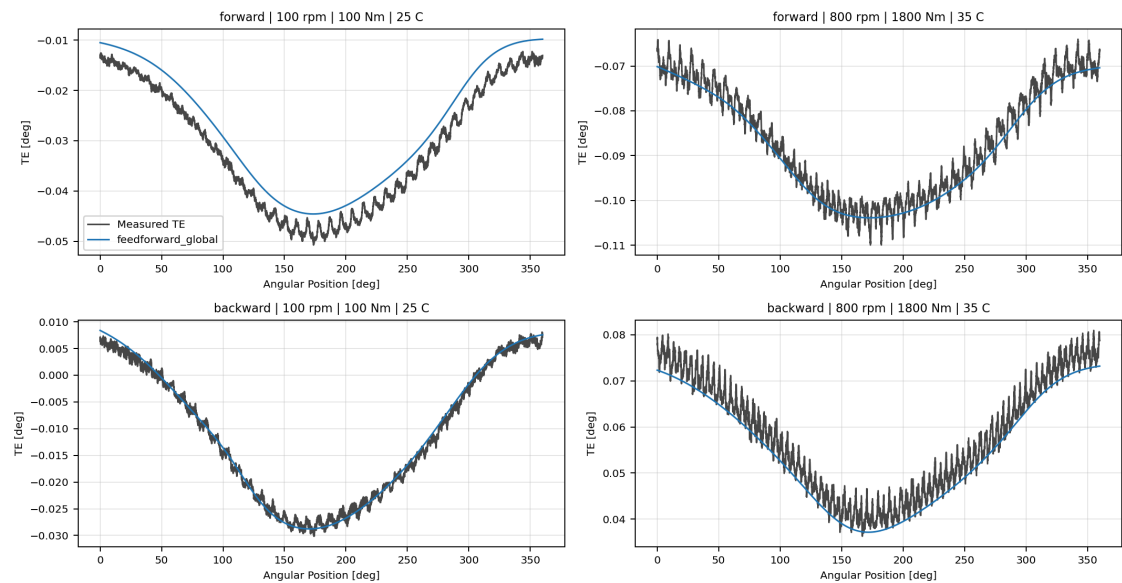


lstm\_sequence\_bw Track 2 collage

# Collage Gallery - Global Wave 1 Family Best Models

## feedforward\_global

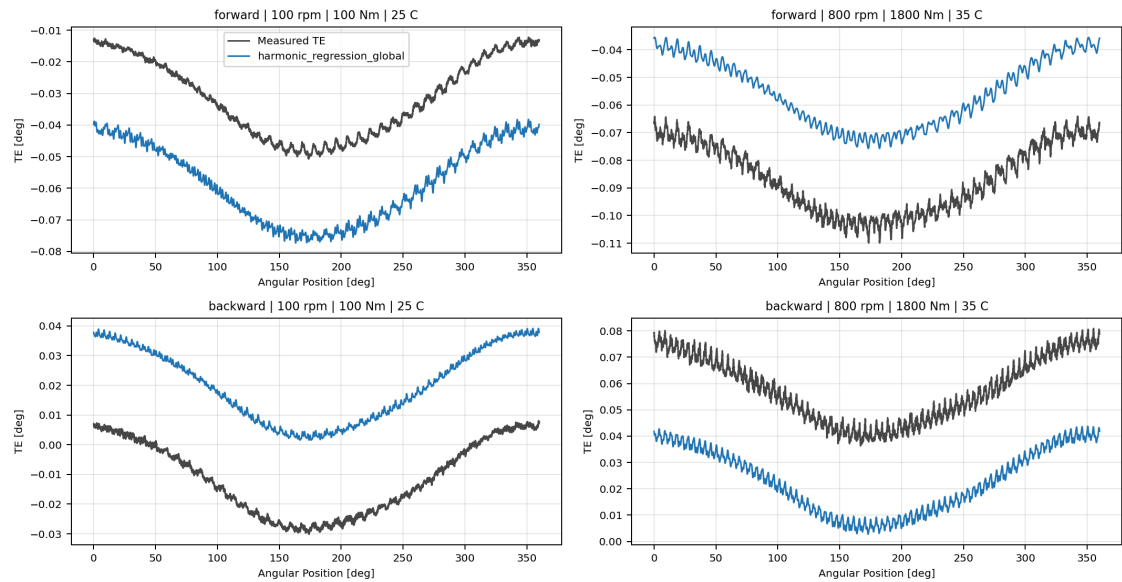
feedforward\_global



feedforward\_global Track 2 collage

## harmonic\_regression\_global

harmonic\_regression\_global

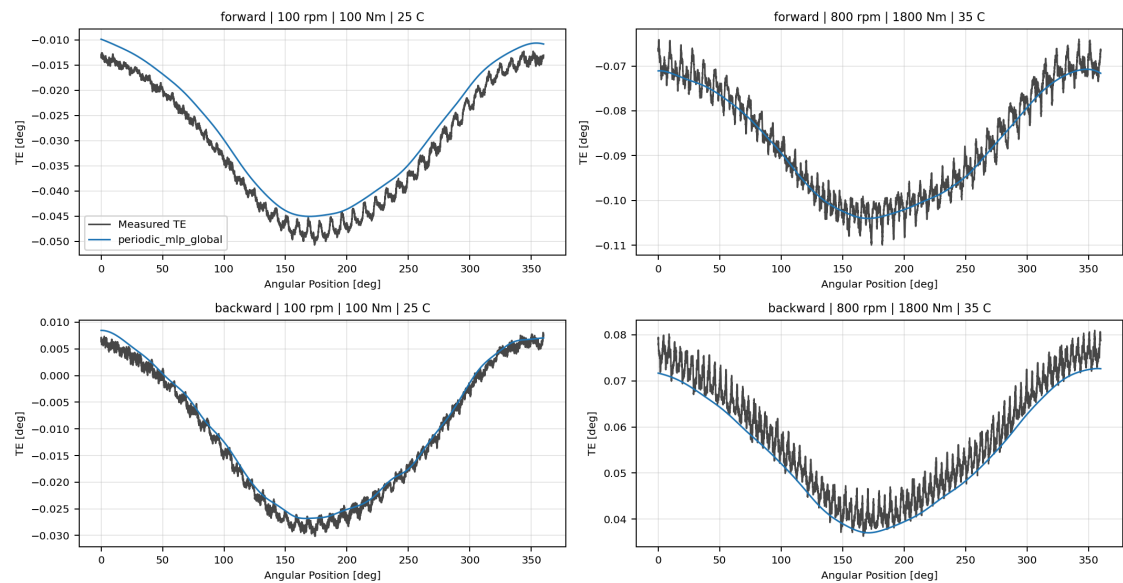


harmonic\_regression\_global Track 2 collage

# Collage Gallery - Global Wave 1 Family Best Models Continued

periodic\_mlp\_global

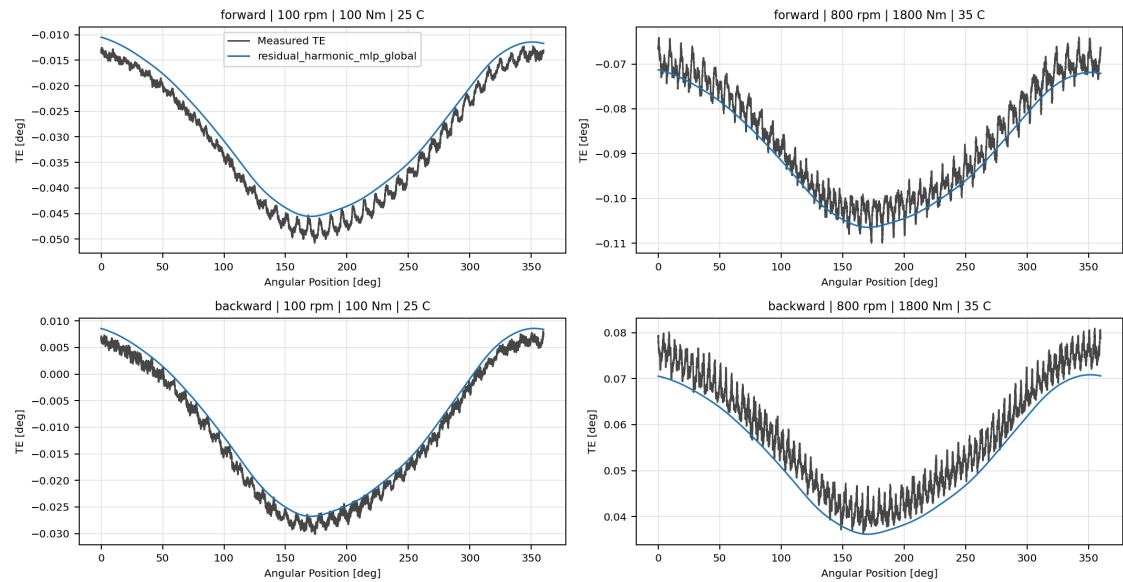
periodic\_mlp\_global



periodic\_mlp\_global Track 2 collage

residual\_harmonic\_mlp\_global

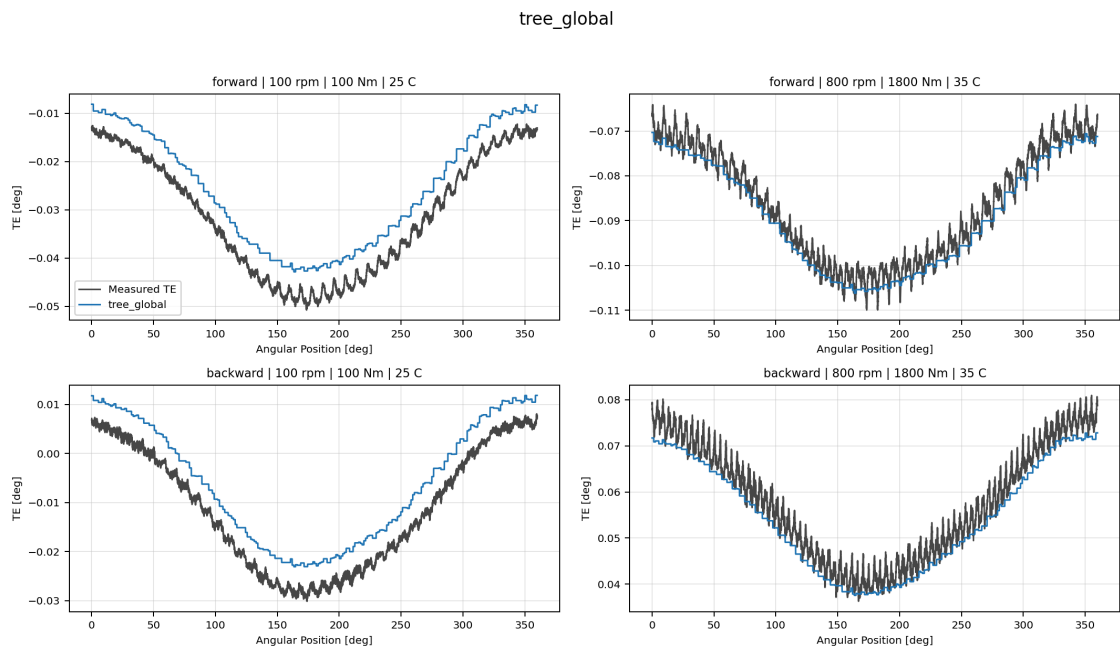
residual\_harmonic\_mlp\_global



residual\_harmonic\_mlp\_global Track 2 collage

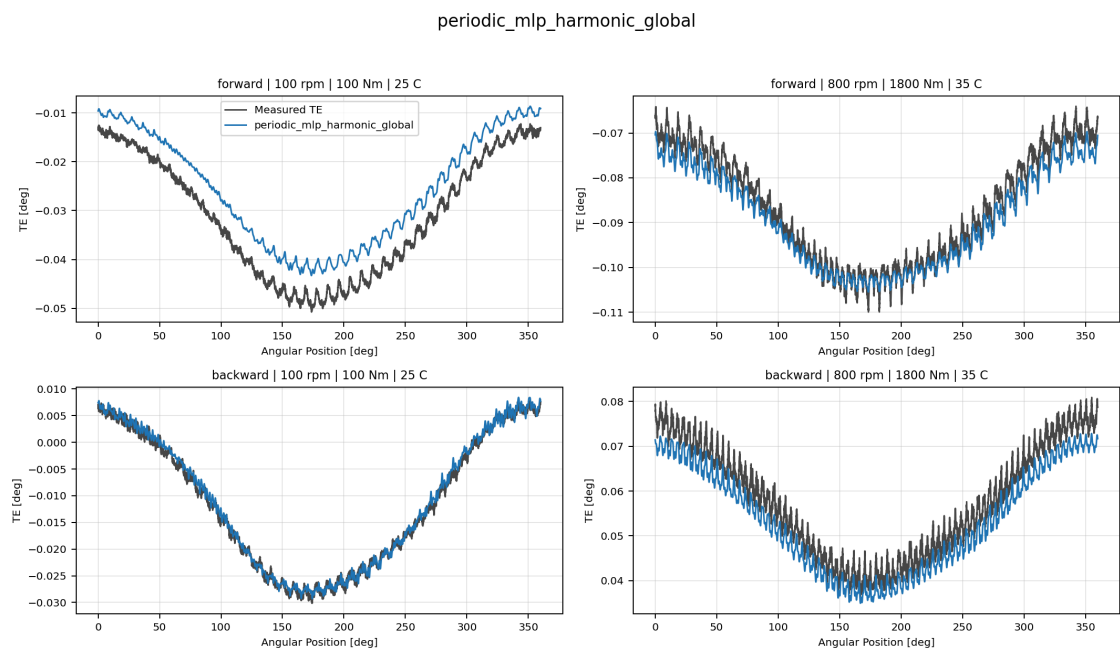
# Collage Gallery - Global Wave 1 Family Best Models Continued 2

tree\_global



tree\_global Track 2 collage

periodic\_mlp\_harmonic\_global

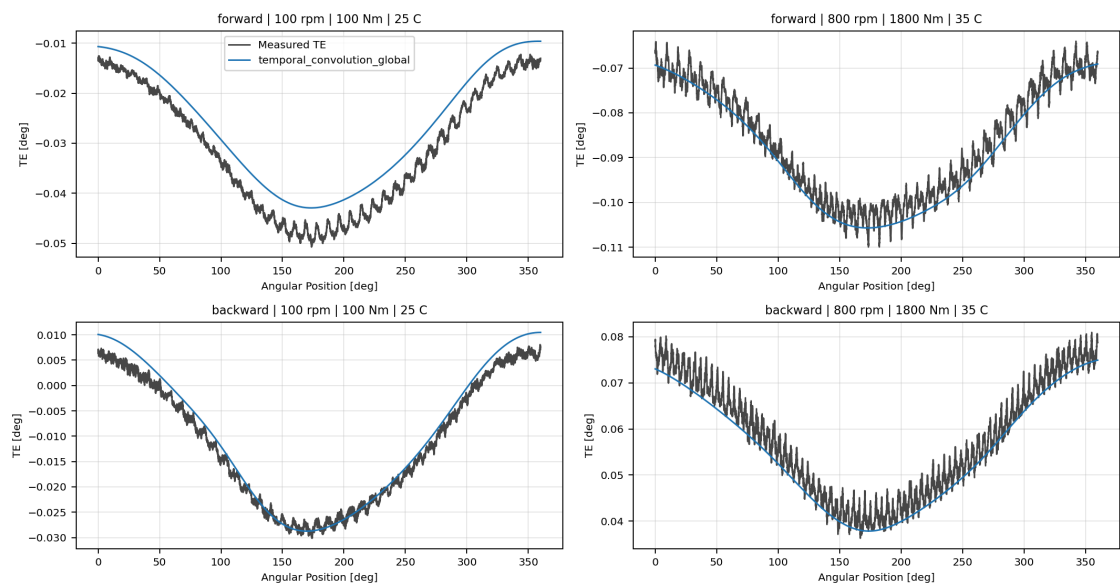


periodic\_mlp\_harmonic\_global Track 2 collage

# Collage Gallery - Global Wave 2 Temporal Family Best Models

temporal\_convolution\_global

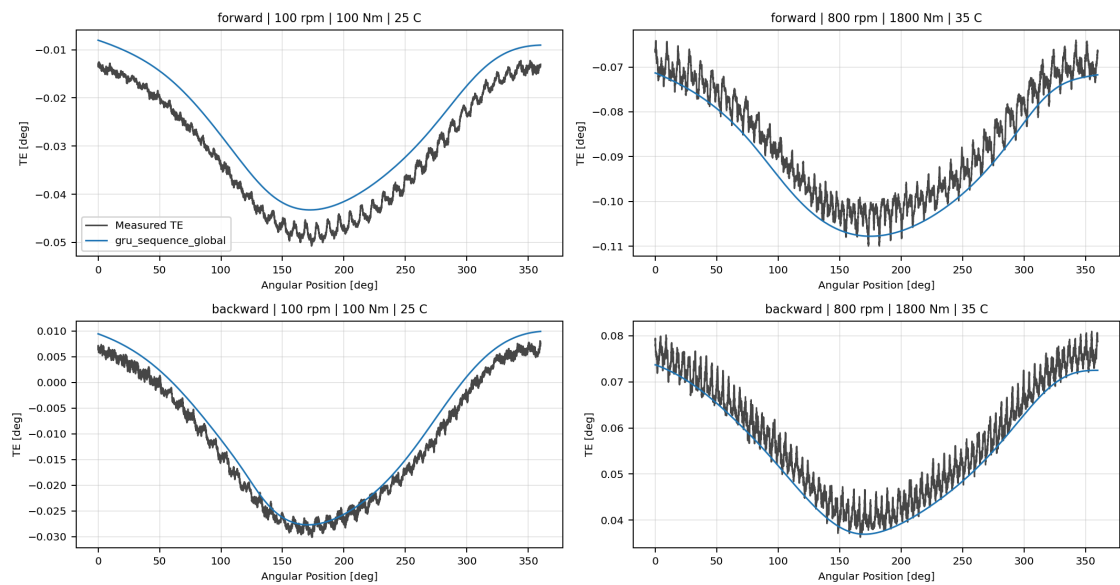
temporal\_convolution\_global



temporal\_convolution\_global Track 2 collage

gru\_sequence\_global

gru\_sequence\_global



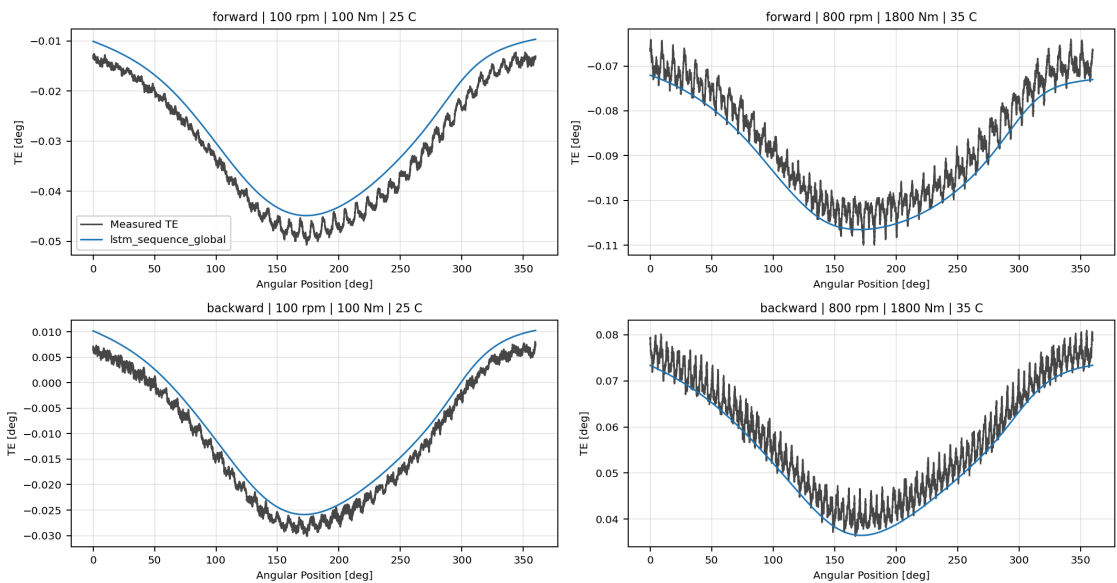
gru\_sequence\_global Track 2 collage



# Collage Gallery - Global Wave 2 Temporal Family Best Models Continued

Istm\_sequence\_global

Istm\_sequence\_global



Istm\_sequence\_global Track 2 collage

## Output Artifacts

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- output directory: `output\validation_checks\track2_best_model_collage_report\2026-05-24-22-50-30__track2_best_model_collage_report` ;
- summary YAML: `output\validation_checks\track2_best_model_collage_report\2026-05-24-22-50-30__track2_best_model_collage_report\track2_best_model_collage_summary.yaml` ;
- metrics CSV: `output\validation_checks\track2_best_model_collage_report\2026-05-24-22-50-30__track2_best_model_collage_report\track2_best_model_collage_metrics.csv` ;
- report Markdown: `doc\reports\analysis\track2\best_model_collage_report\[2026-05-24]\track2_best_model_collage_report.md` .